

Lecture #2

Content to be covered

- ❑ Preview of superposition
- ❑ Generation of Coherent sources
 - ❖ Division of wave front
 - ❖ Brief about YDSE
- ❑ Fresnel's bi-prism
 - ❖ Distance Between Virtual Sources: Displacement method
 - ❖ Distance Between Virtual Sources: Deviation method
 - ❖ Displacement of fringes
 - ❖ Fringes with white light

Resources to be consulted

- ❑ OPTICS- by Ajay Ghatak
- ❑ <https://youtu.be/hvpYKPyT-vc>

Superposition of waves

Consider the superposition of two sinusoidal waves at a particular point. Let the displacements produced by these waves are given as

$$\begin{aligned}y_1 &= a_1 \sin(\omega t + \phi_1) \\y_2 &= a_2 \sin(\omega t + \phi_2)\end{aligned}$$

Resultant displacement according to superposition principle

$$y = y_1 + y_2 = a_1 \sin(\omega t + \phi_1) + a_2 \sin(\omega t + \phi_2)$$

$$\text{Let, } a_1 \cos \phi_1 + a_2 \cos \phi_2 = A \cos \alpha \quad \text{and}$$

$$a_1 \sin \phi_1 + a_2 \sin \phi_2 = A \sin \alpha$$

$$\Rightarrow y = A \cos \alpha \sin \omega t + A \sin \alpha \cos \omega t$$

$$\Rightarrow y = A \sin(\omega t + \alpha)$$

&

$$\tan \alpha = \frac{a_1 \sin \phi_1 + a_2 \sin \phi_2}{a_1 \cos \phi_1 + a_2 \cos \phi_2}$$

$$I = A^2 = a_1^2 + a_2^2 + 2a_1a_2 \cos(\phi_2 - \phi_1)$$

$$\Rightarrow I = A^2 = a_1^2 + a_2^2 + 2a_1a_2 \cos \phi$$

$\phi = (\phi_2 - \phi_1)$
=Phase difference

$$y_1 = a_1 \sin(\omega t + \phi_1) \quad \text{--- (1)}$$

$$y_2 = a_2 \sin(\omega t + \phi_2) \quad \text{--- (2)}$$

Resultant displacement according to Superposition principle

$$y = y_1 + y_2 = a_1 \sin(\omega t + \phi_1) + a_2 \sin(\omega t + \phi_2)$$

$$\Rightarrow y = (a_1 \cos \phi_1 + a_2 \cos \phi_2) \sin \omega t + \underbrace{(a_1 \sin \phi_1 + a_2 \sin \phi_2)}_{\cos \omega t}$$

$$\text{Let } \left. \begin{aligned} a_1 \cos \phi_1 + a_2 \cos \phi_2 &= A \cos \alpha \\ \text{and} \end{aligned} \right\} \text{--- (3)}$$

$$a_1 \sin \phi_1 + a_2 \sin \phi_2 = A \sin \alpha$$

$$\Rightarrow y = A \cos \alpha \sin \omega t + A \sin \alpha \cos \omega t$$

$$\Rightarrow y = A \sin(\omega t + \alpha)$$

Squaring and adding (1)

$$a_1^2 \cos^2 \phi_1 + a_2^2 \cos^2 \phi_2 + 2a_1 a_2 \cos \phi_1 \cos \phi_2$$

$$+ a_1^2 \sin^2 \phi_1 + a_2^2 \sin^2 \phi_2 + 2a_1 a_2 \sin \phi_1 \sin \phi_2 = r^2$$

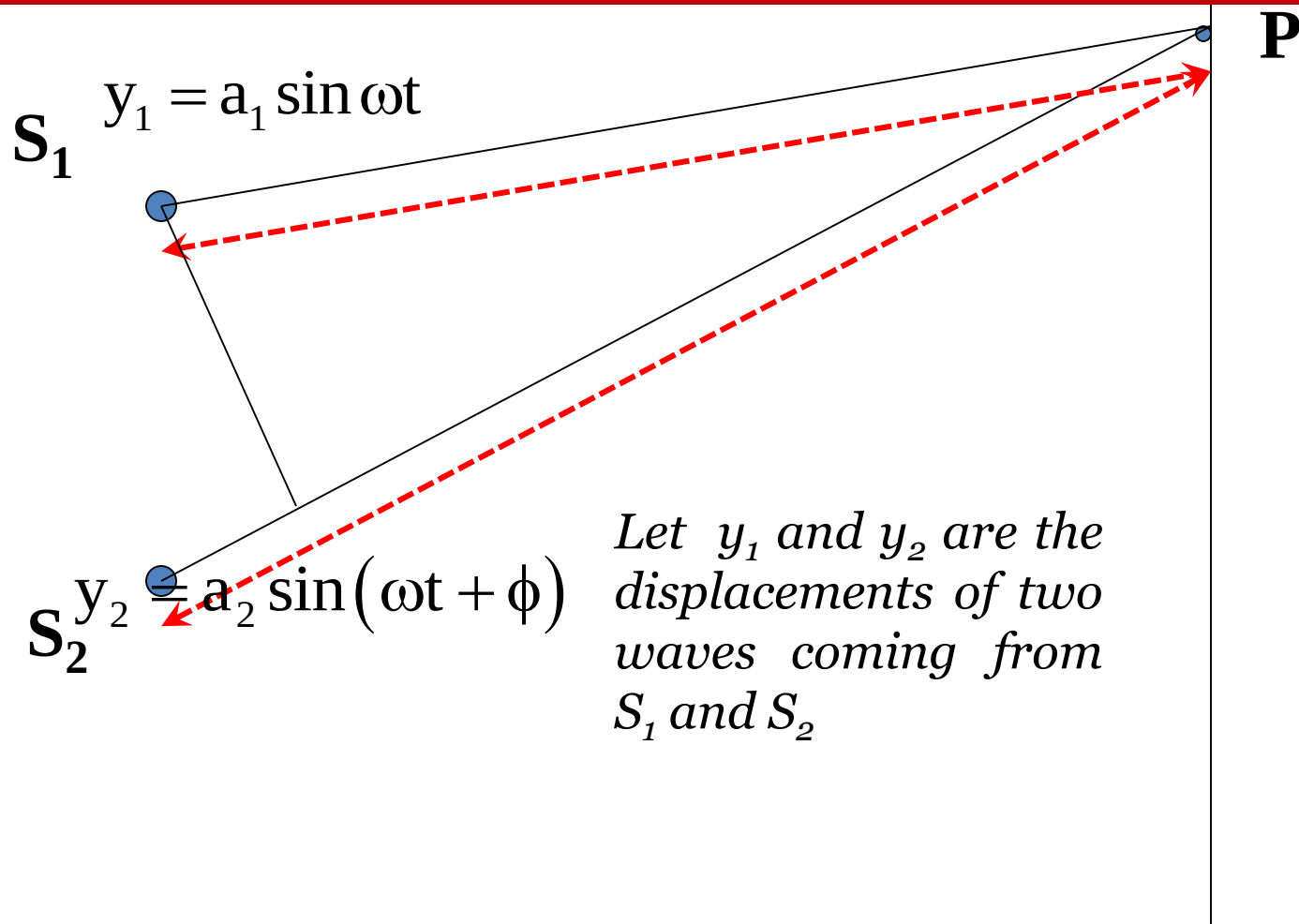
$$I = A^2 = a_1^2 + a_2^2 + 2a_1 a_2 \cos(\phi_2 - \phi_1) \quad \left(\begin{array}{l} \cos(A-B) \\ = \cos A \cos B \\ + \sin A \sin B \end{array} \right)$$

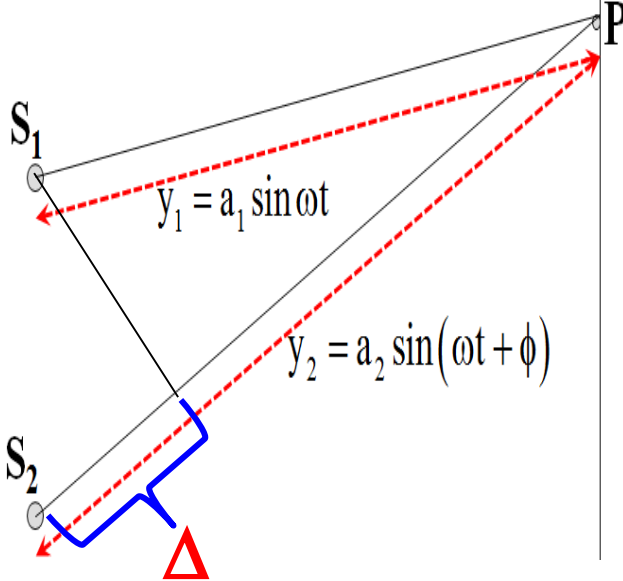
$$\Rightarrow I = A^2 = a_1^2 + a_2^2 + 2a_1 a_2 \cos \phi \quad (4)$$

$$\phi = \phi_2 - \phi_1 = \text{Phase difference}$$

Conclusion: Since amplitudes are constant therefore intensity varies with phase difference

Superposition of two waves: Intensity distribution





$$\begin{aligned}
 y &= y_1 + y_2 = a_1 \sin \omega t + a_2 \sin(\omega t + \phi) \\
 &= a_1 \sin \omega t + a_2 \cos \phi \sin \omega t + a_2 \sin \phi \cos \omega t \\
 &= \underline{(a_1 + a_2 \cos \phi)} \sin \omega t + \underline{\underline{a_2 \sin \phi}} \cos \omega t
 \end{aligned}$$

$$\begin{aligned}
 y &= \underline{A \cos \delta} \sin \omega t + \underline{\underline{A \sin \delta}} \cos \omega t \\
 &= A \sin(\omega t + \delta)
 \end{aligned}$$

Resultant Intensity = (Resultant Amplitude)²

$$(a_1 + a_2 \cos \phi)^2 + (a_2 \sin \phi)^2 = A^2 = I$$

$$I = a_1^2 + a_2^2 + 2a_1a_2 \underline{\cos \phi}$$

$$I = (a_1 - a_2)^2 + 4a_1a_2 \cos^2 \phi / 2$$

$I \propto \text{cosine square}$

MAXIMA

$$I = (a_1 - a_2)^2 + 4a_1a_2 \cos^2 \phi / 2$$

when $a_1 = a_2$

$$I = 4a^2 \cos^2 \phi / 2$$

if $\phi = 0, \pm 2\pi, \pm 4\pi, \dots, 2n\pi$ $n=0,1,2,\dots$

$$I_{\max} = (a_1 + a_2)^2 \Rightarrow I_{\max} > I_1 + I_2$$

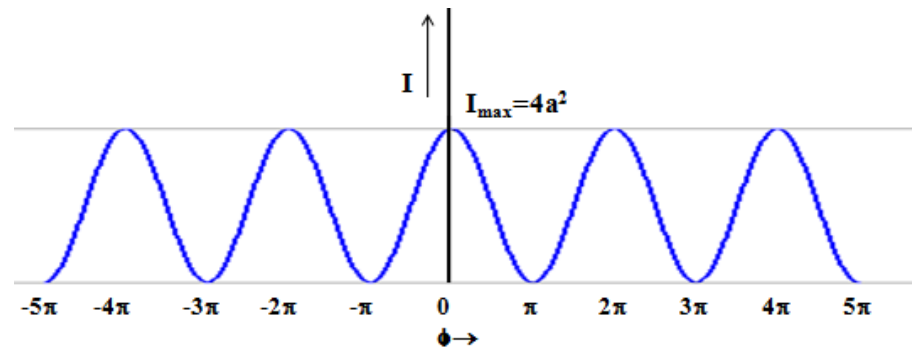
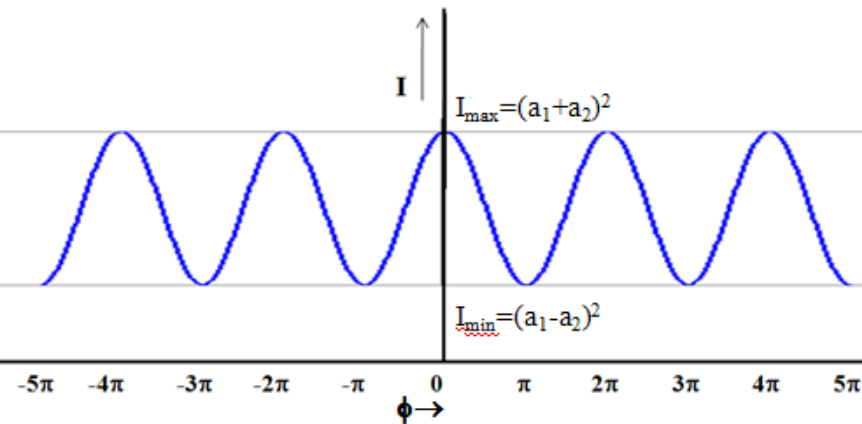
MINIMA

if $\phi = \pm\pi, \pm 3\pi, \dots, (2n-1)\pi$ $n=1,2,3,\dots$

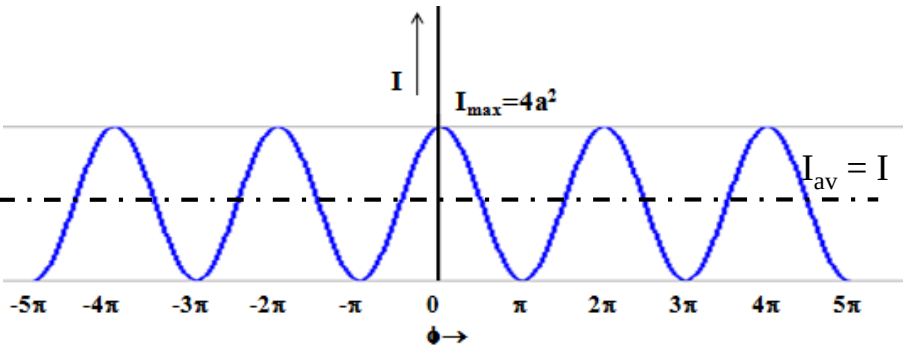
$$I_{\min} = (a_1 - a_2)^2 \Rightarrow I_{\min} < I_1 + I_2$$

$$I_{\max} = 4a^2$$

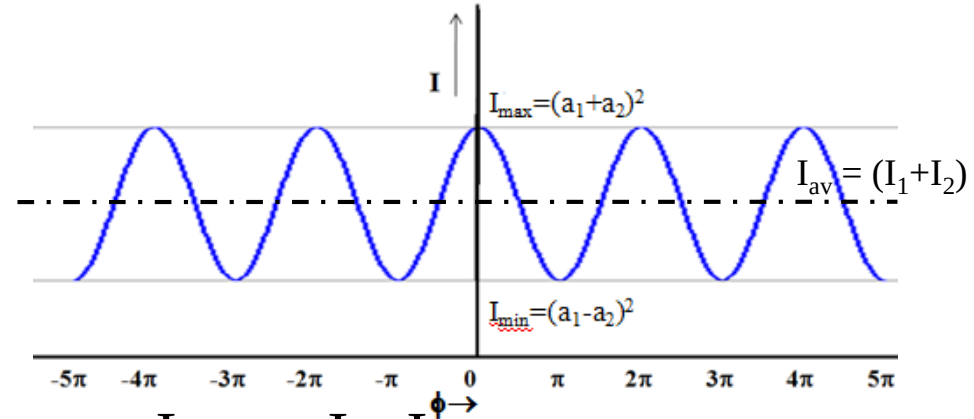
$$I_{\min} = 0$$



Law of conservation of energy



$$\Rightarrow I_{\max} > I_1 + I_2$$



$$\Rightarrow I_{\min} < I_1 + I_2$$

$$\begin{aligned} I_{\text{av}} &= \frac{\int_0^{2\pi} I d\phi}{\int_0^{2\pi} d\phi} = \frac{\int_0^{2\pi} (a_1^2 + a_2^2 + 2a_1a_2 \cos \phi) d\phi}{\int_0^{2\pi} d\phi} \\ &= \frac{(a_1^2 + a_2^2)2\pi}{2\pi} = a_1^2 + a_2^2 \\ &= I_1 + I_2 \end{aligned}$$

This establishes that in the interference pattern, intensity of light is simply being redistributed i.e. energy is being transferred from regions of destructive interference to the regions of constructive interference. No energy is being created or destroyed in the process. Thus the principle of energy conservation is being obeyed in the process of interference of light

For incoherent light

$$I = (a_1 - a_2)^2 + 4a_1a_2 \cos^2 \phi / 2$$

$$I = (a_1 - a_2)^2 + 4a_1a_2 \left\langle \cos^2 \phi / 2 \right\rangle$$

$$= (a_1 - a_2)^2 + 4a_1a_2 \times \frac{1}{2} = (a_1 - a_2)^2 + 2a_1a_2$$

$$= a_1^2 + a_2^2$$

$$= I_1 + I_2$$

= sum of intensity
of constituent waves.

If amplitudes and phases are different

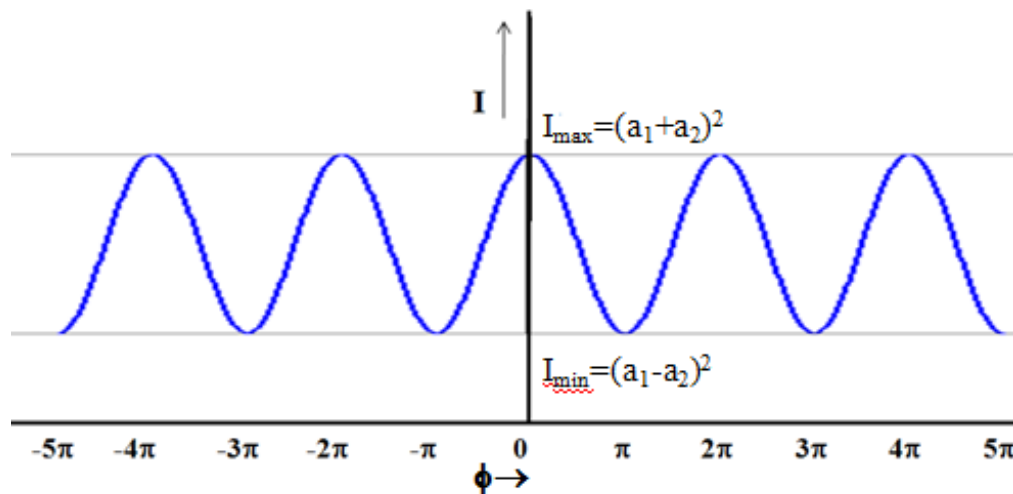
$$y_1 = a_1 \cos(\omega t)$$

$$y_2 = a_2 \cos(\omega t + \phi)$$

→
$$I = a_1^2 + a_2^2 + 2a_1a_2 \cos(\phi)$$

$$I_{\max} = ? \quad I_{\min} = ?$$

Visibility of the fringes (V)



$$V \equiv \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}}$$

THE INTERFERENCE FRINGES

$$(S_2P)^2 = [D^2 + (y_n + \frac{d}{2})^2]$$

$$(S_1P)^2 = [D^2 + (y_n - \frac{d}{2})^2]$$

$$(S_2P)^2 - (S_1P)^2 = 2y_nd$$

$$S_2P - S_1P = \frac{2y_nd}{S_2P + S_1P}$$

If $d \ll D$ then $S_2P \approx S_1P \approx D$

Path difference

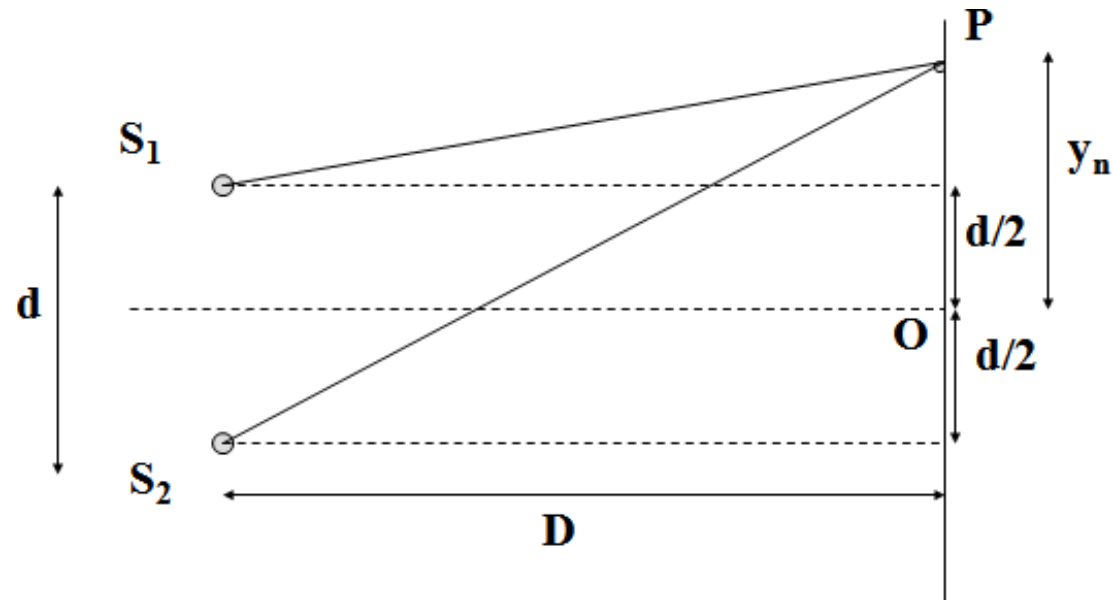
$$S_2P - S_1P = \frac{y_nd}{D}$$

At point P for maxima we must have

$$S_2P - S_1P = n\lambda, \quad n = 0, 1, 2, 3, \dots$$

$$\text{therefore, } n\lambda = \frac{2y_nd}{2D} = \frac{y_nd}{D}$$

$$\text{or, } n\lambda = \frac{y_nd}{D}$$



Production of Coherent Sources

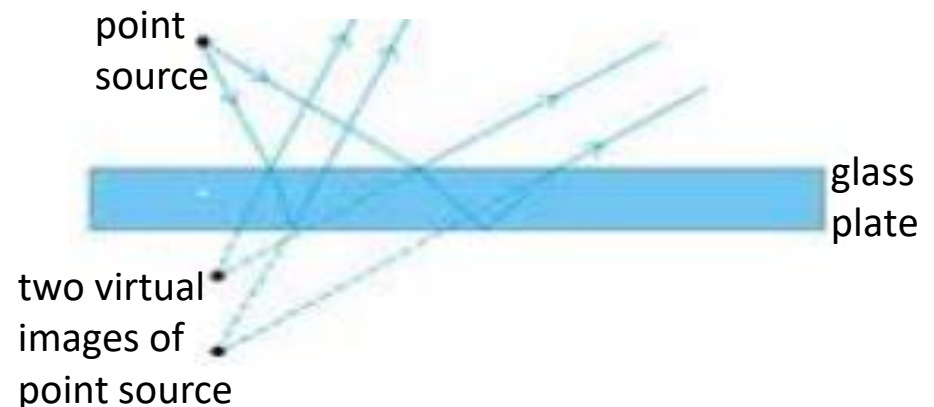
In practice it is not possible to get independent two sources which are coherent. However, two virtual sources, formed from one single source, can act as coherent sources. There are two ways to get two coherent sources from one single source

1. By Division of Wave front: Two wave fronts are produced from a single wave

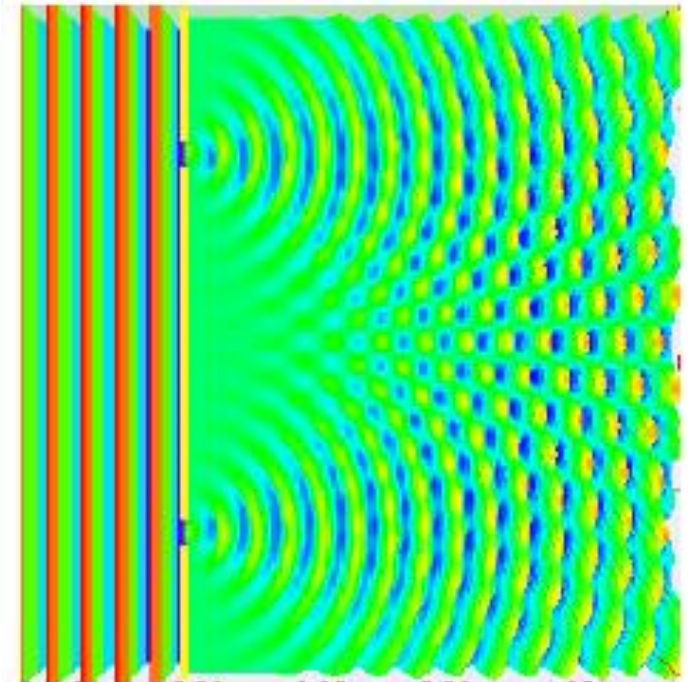
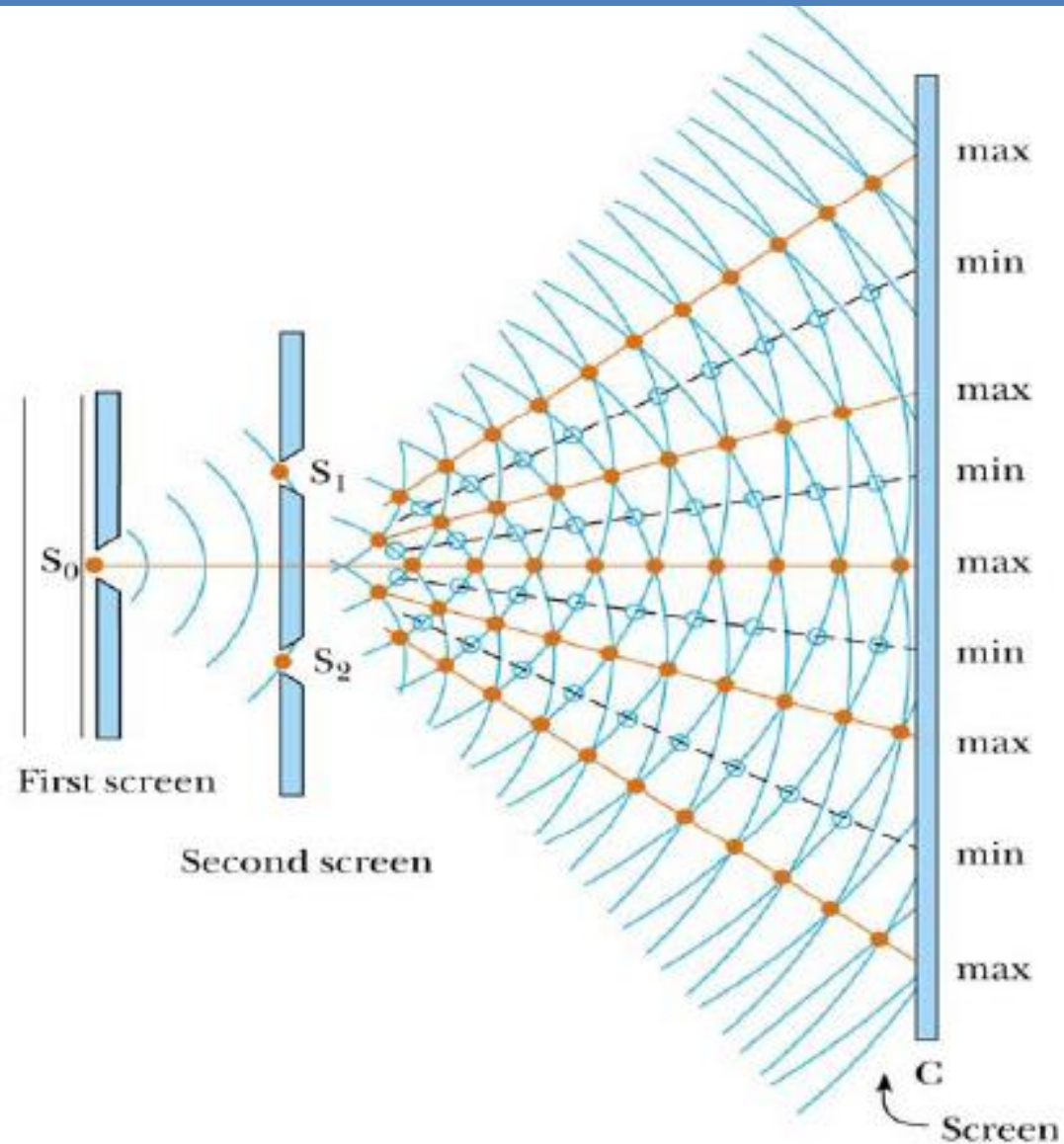
Examples: Young's double slit experiment,
Fresnel Bi-prism, Lloyd's Mirror, Fresnel Mirror

2. By Division of Amplitude: Two waves are formed from a single source due to reflection and refraction. One of these source is real and other is virtual source.

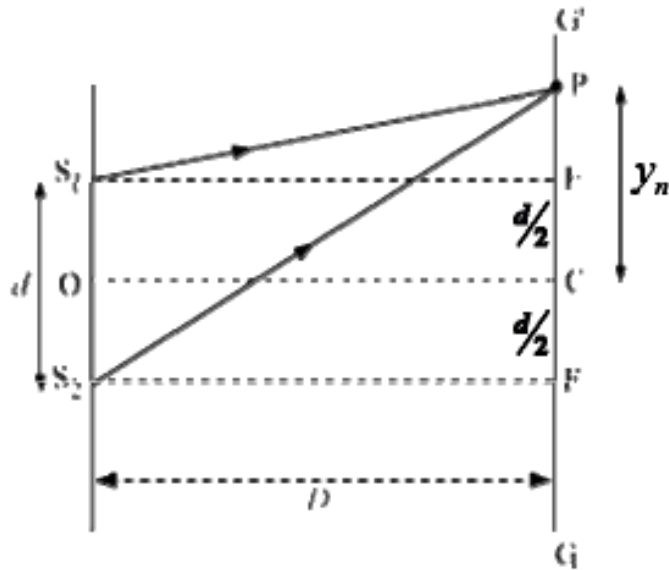
Examples: Newton's Ring,
Michelson Interferometer,
Thin film interference



Interference Fringes : Division of wave front



Interference Fringes : Division of wave front



For Intensity to be maximum at Point P
 $S_2P - S_1P = n\lambda$

If $d \ll D$, then $S_2P \approx S_1P \approx D$, we will get

$$\frac{y_n d}{D} = n\lambda \quad \text{Bright Fringe}$$

Distance between any two consecutive bright fringes

$$y_2 - y_1 = y_3 - y_2$$

$$= \frac{2\lambda D}{d} - \frac{\lambda D}{d} = \frac{3\lambda D}{d} - \frac{2\lambda D}{d}$$

$$\beta = \frac{\lambda D}{d} \quad \text{Fringe width}$$

$$n = 0 \Rightarrow y_0 = 0 \quad (0^{\text{th}} \text{ order central fringe})$$

$$n = 1 \Rightarrow y_1 = \frac{\lambda D}{d} \quad (1^{\text{st}} \text{ order fringe})$$

$$n = 2 \Rightarrow y_2 = \frac{2\lambda D}{d} \quad (2^{\text{nd}} \text{ order fringe})$$

$$n = 3 \Rightarrow y_3 = \frac{3\lambda D}{d} \quad (3^{\text{rd}} \text{ order fringe})$$

Interference Fringes : Division of wave front

Similarly for dark fringes,

$$S_2P - S_1P = (2n+1)\frac{\lambda}{2}, n = 0, 1, 2, 3..$$

So we get, for dark and bright fringes

$$y_n = \frac{(2n+1)\lambda D}{2d} \quad \text{Dark Fringe}$$

$$n = 0, \rightarrow y_0 = \frac{\lambda D}{2d}$$

$$n = 1, \rightarrow y_1 = \frac{3\lambda D}{2d}$$

$$n = 2, \rightarrow y_2 = \frac{5\lambda D}{2d}$$

$$y_1 - y_0 = y_2 - y_1$$

$$= \frac{5\lambda D}{2d} - \frac{3\lambda D}{2d} = \frac{\lambda D}{d}$$

$$\Rightarrow \beta = \frac{\lambda D}{d}$$

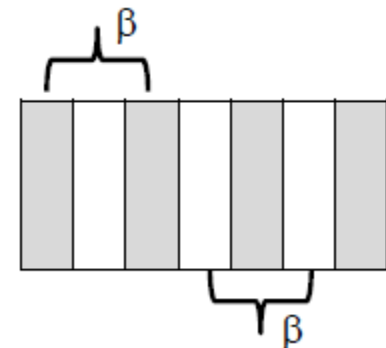
Thus fringe width of dark and bright fringes are same.

Separation between dark and bright fringes

$$\frac{3\lambda D}{2d} - \frac{\lambda D}{d} = \frac{\lambda D}{2d} = \frac{\beta}{2}$$

Visibility of the fringes (V)

$$V \equiv \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}}$$



Two beams of equal irradiance (I_0): case study YDSE

Intensity is given by

$$I = 4I_0 \cos^2 \frac{\phi}{2}$$

But path difference for the case of YDSE

$$\Delta = \frac{yd}{D}$$

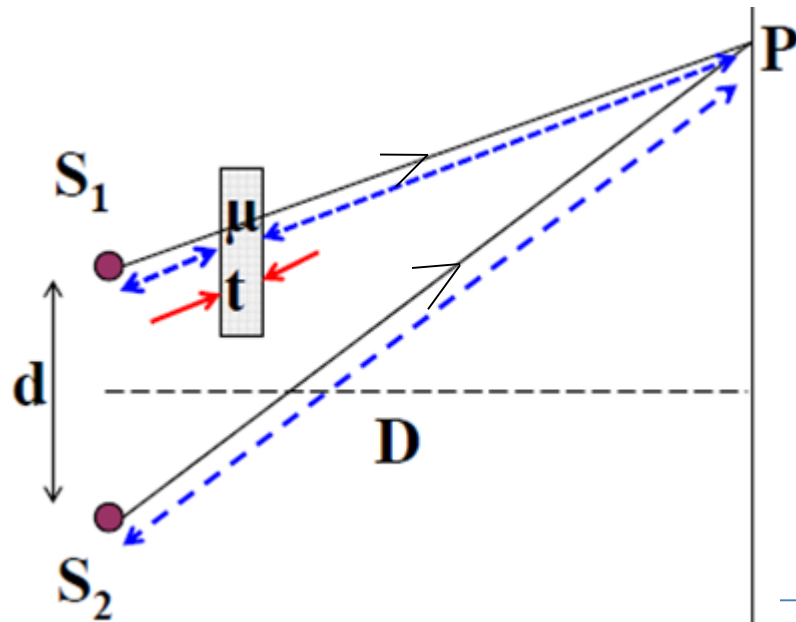
the phase difference is given by

$$\phi = \frac{2\pi}{\lambda} \frac{yd}{D}$$

Hence, the maximum intensity

$$I = 4I_0 \cos^2 \frac{\pi yd}{\lambda D}$$

In Double slit experiment: Insert a thin transparent glass sheet of thickness t and refractive index μ in the path of one beam.



Time required for the light to reach from S_1 to the point P is

$$T = \frac{S_1P - t}{c} + \frac{t}{v} = \frac{S_1P + (\mu - 1)t}{c}$$

where 'c' is the velocity of light in air and 'v' its velocity in the medium

since $\frac{c}{v} = \mu \rightarrow$ refractive index

Time required for the light to reach from S_2 to the point P is

$$T' = \frac{S_2P}{c}$$

To interfere at point P, they should reach there at same time i.e. $T = T'$.

Path difference

$$\Delta = S_2P - S_1P - (\mu - 1)t$$

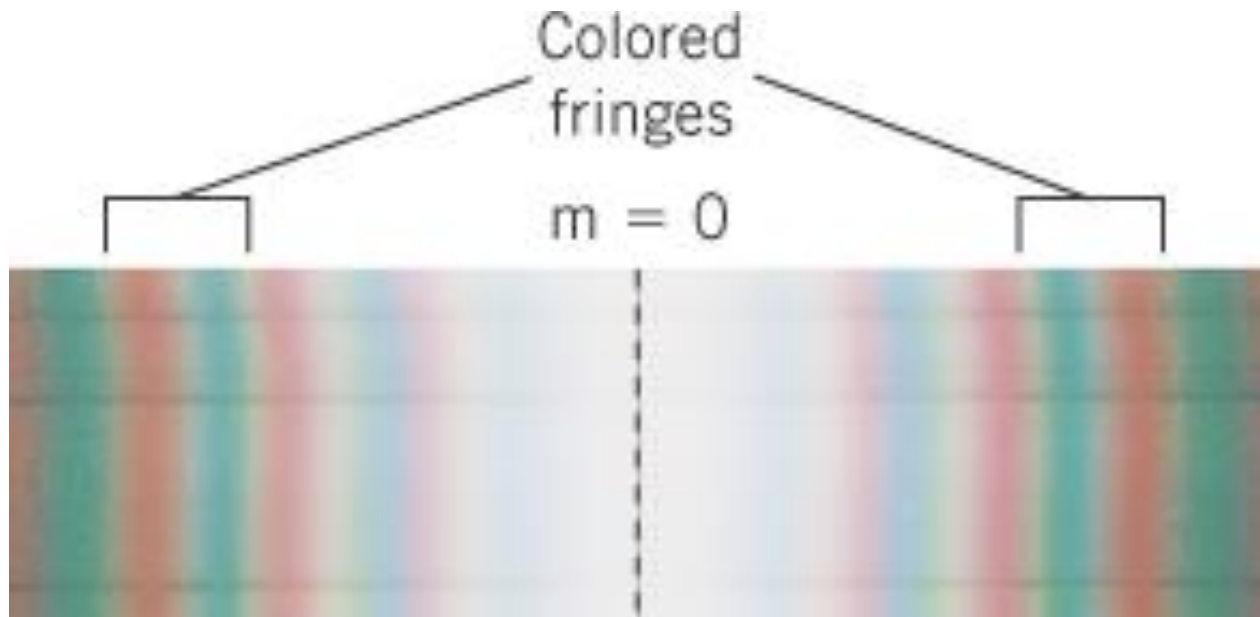
$$\Delta = \frac{y_n d}{D} - (\mu - 1)t$$

position of n^{th} bright fringe

$$y_n = \frac{D}{d} [n\lambda + (\mu - 1)t]$$

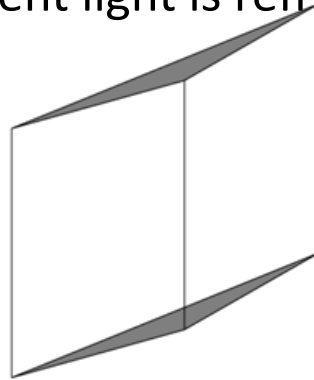
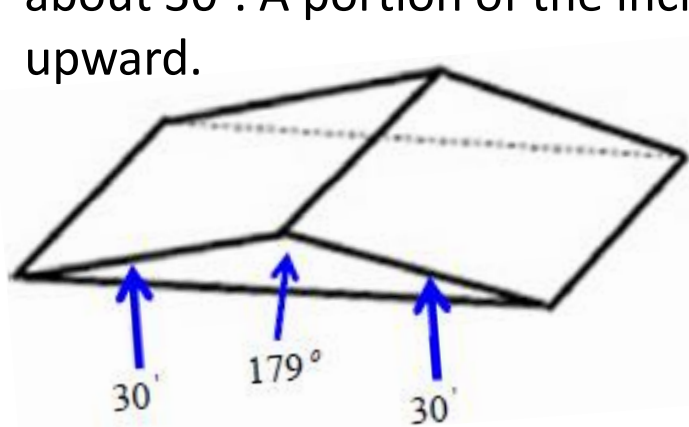
YDSE with White Light

The figure shows a photograph that illustrates the kind of interference fringes that can result when white light is used in Young's experiment. Why does Young's experiment separate white light into its constituent colors? In any group of colored fringes, such as the two singled out, why is red farther out from the central fringe than green is? Why is the central fringe white?



Fresnel Biprism: (Augustin-Jean Fresnel)

It consists of two thin acute angled prisms joined at the bases. It is constructed as a single prism of obtuse angle of 179° . The acute angle α on both side is about $30'$. A portion of the incident light is refracted downward and a portion upward.



The fringe width

$$\beta = \lambda D / d$$

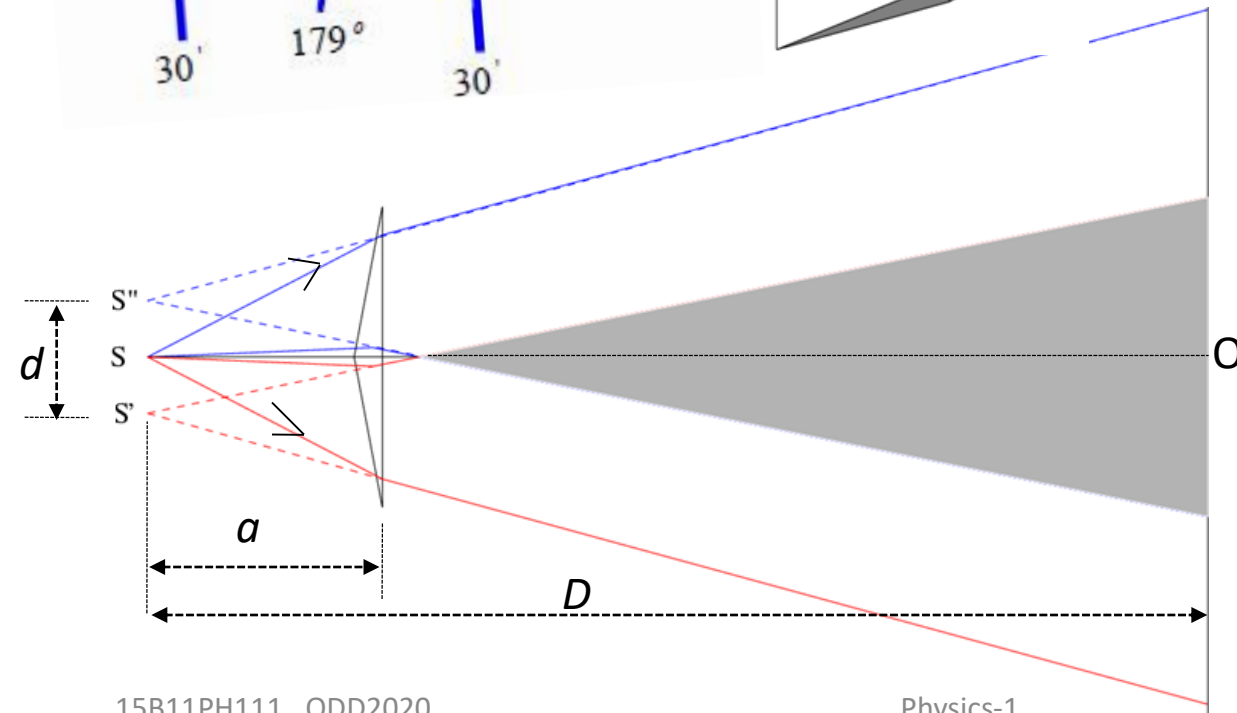
Position of bright fringes from O

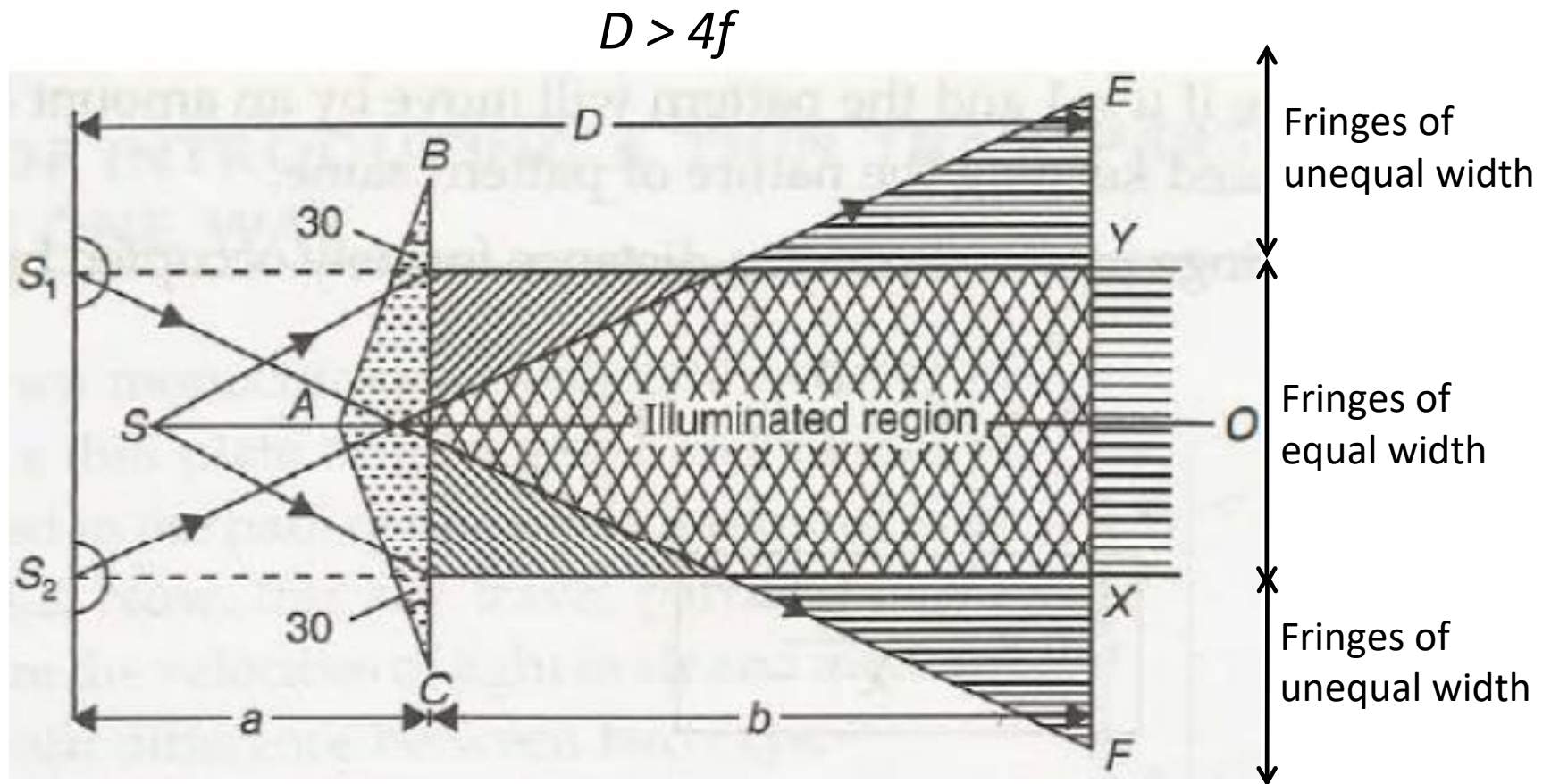
$$y_n = \frac{n\lambda D}{d}$$

Position of dark fringes from O

$$y_n = \frac{(2n+1)\lambda D}{2d}$$

where $n = 0, 1, 2, 3, \dots$

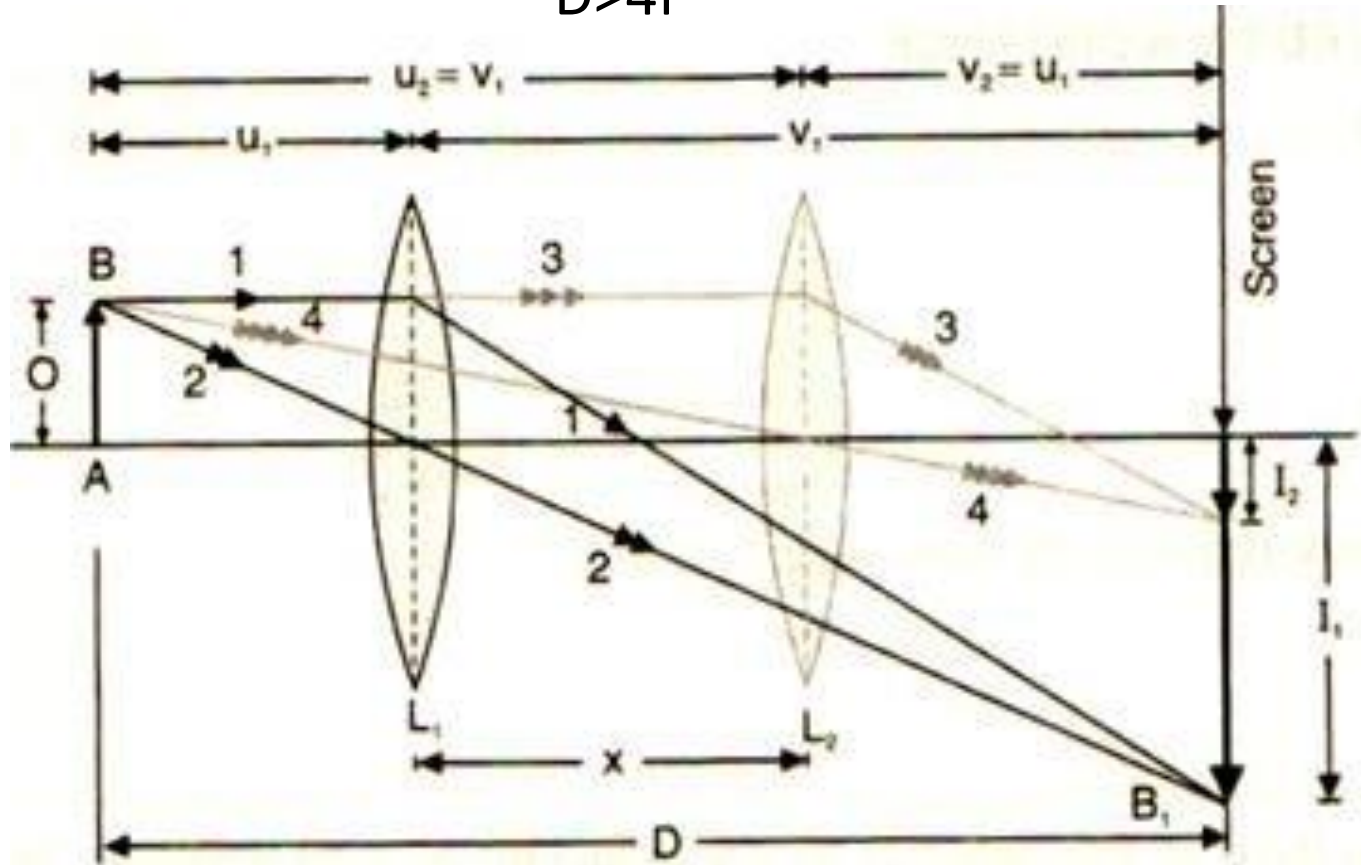




Angular fringe width $\Delta\theta = \frac{\beta}{D} = \frac{\lambda}{d}$

Distance Between Virtual Sources: Displacement method

$D > 4f$



$$\frac{I_1}{O} = \frac{v_1}{u_1}$$

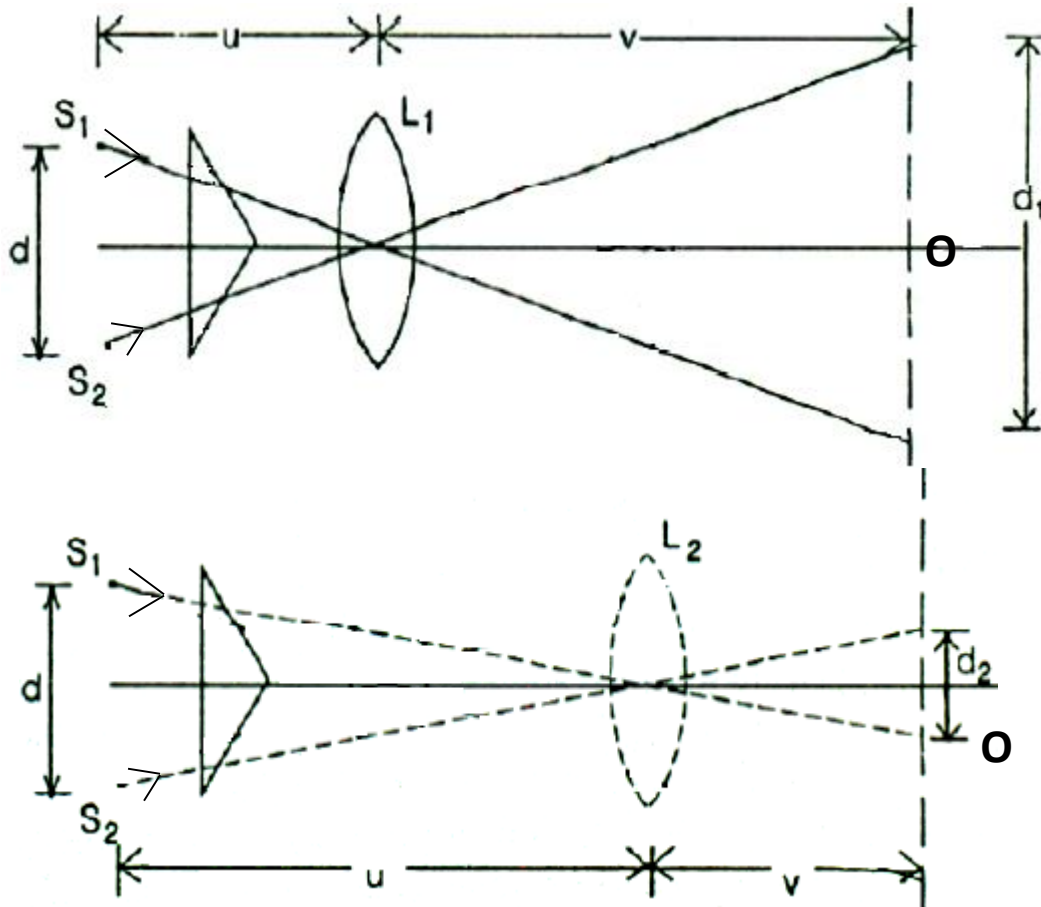
and

$$\frac{I_2}{O} = \frac{v_2}{u_2}$$

But $u_2 = v_1$
 $v_2 = u_1$

$$O = \sqrt{I_1 I_2}$$

Distance Between Virtual Sources: Displacement method



$$d = \sqrt{d_1 d_2}$$

Now the fringe width is given by

$$\beta = \frac{\lambda D}{\sqrt{d_1 d_2}}$$

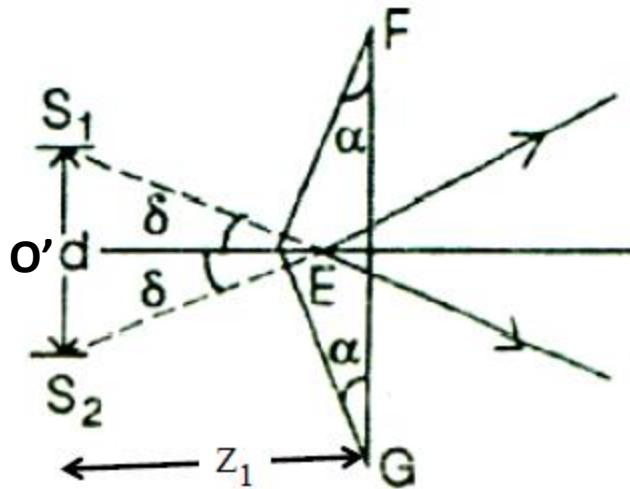
and wavelength

$$\lambda = \frac{\beta \sqrt{d_1 d_2}}{D}$$

$$\beta = \frac{L}{20} \quad \uparrow$$

Eyepiece is moved horizontally to determine the fringe width. Suppose 20 bright fringes cross the field of view, when the eyepiece moves through a distance L , then

Distance Between Virtual Sources: Deviation method



For small angles $\delta = (\mu - 1)\alpha$ (1)

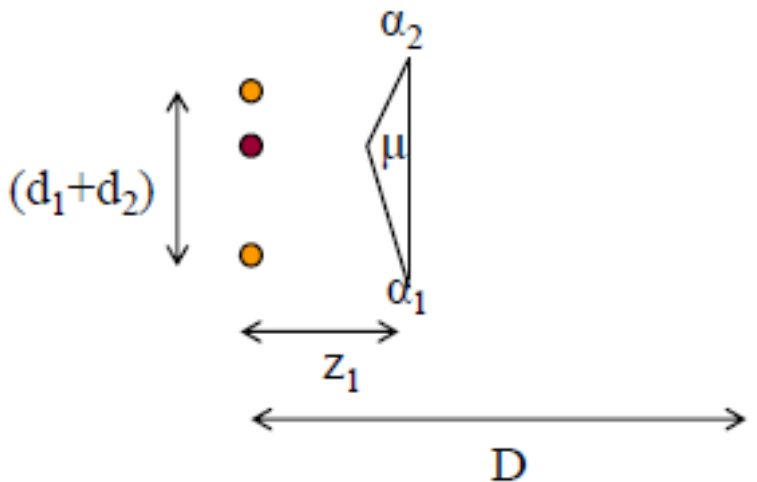
δ is angle of deviation

From right angle triangle S_1EO' and equation (1)

$$\delta = \frac{d/2}{z_1}$$

Hence,

$$d = 2\alpha(\mu - 1)z_1$$



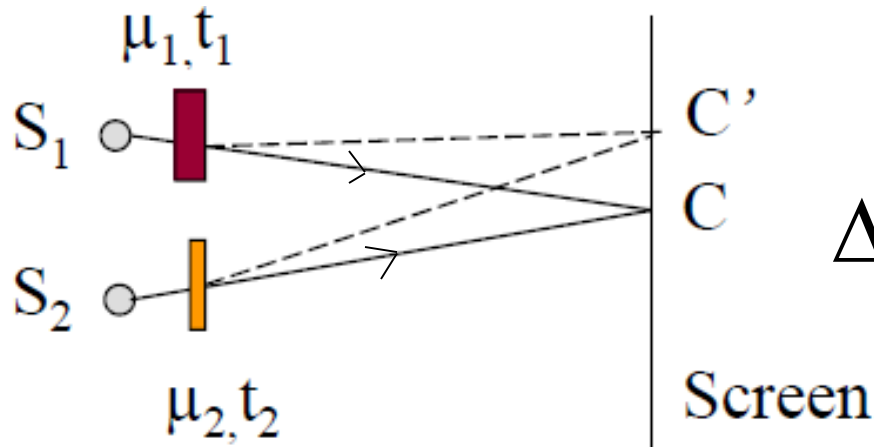
If base angles are different, then

$$d = d_1 + d_2 = (\mu - 1)(\alpha_1 + \alpha_2)z_1$$

$$\beta = \frac{\lambda D}{d_1 + d_2} = \frac{\lambda D}{(\mu - 1)(\alpha_1 + \alpha_2)z_1}$$

Fringe Displacement

Determine condition of net path difference



$$\Delta = (\mu_1 - 1)t_1 - (\mu_2 - 1)t_2$$

Case 1:

If $\mu_1 = \mu_2 = \mu$ and $t_1 > t_2$, Δ is positive (upward shift)
or $t_2 > t_1$, Δ is negative (downward shift)

Case 2:

If $t_1 = t_2 = t$ and $\mu_1 > \mu_2$, Δ is positive (upward shift)
or $\mu_2 > \mu_1$, Δ is negative (downward shift)

Fringes with White Light

When white light is used the center fringe at O is white since all waves will constructively interfere here while the fringes on the both side of O are colored because the fringe width (β) depends on wavelength of light.

For green light $y_g = \frac{n\lambda_g D}{2(\mu_g - 1)\alpha z_1}$

For red light $y_r = \frac{n\lambda_r D}{2(\mu_r - 1)\alpha z_1}$



White light is used to determine shift of fringe pattern, and thickness of the material placed in the path of interfering beam.